Kilonovae and *R*-process Nucleosynthesis from Neutron Star Mergers

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NC STATE UNIVERSITY

LDRD

NS merger illustration: NASA Goddard

Outline

1. Introduction
   - Question: what is the astrophysical origin of *heavy* elements?
     - Answer: Neutron star mergers?
2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)
3. Modelling Astrophysics: How much *r*-process is produced? (mass)
4. Conclusions

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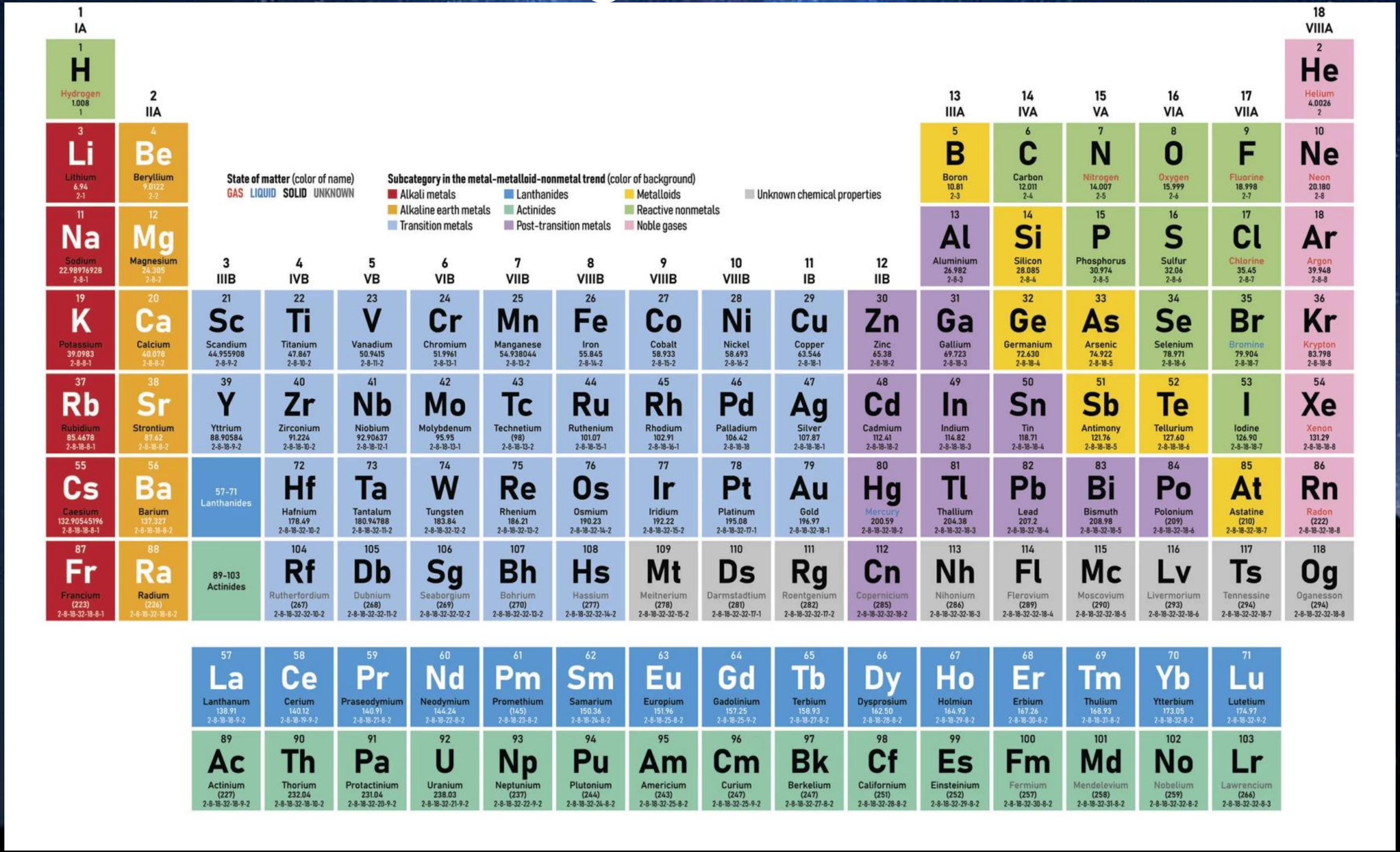
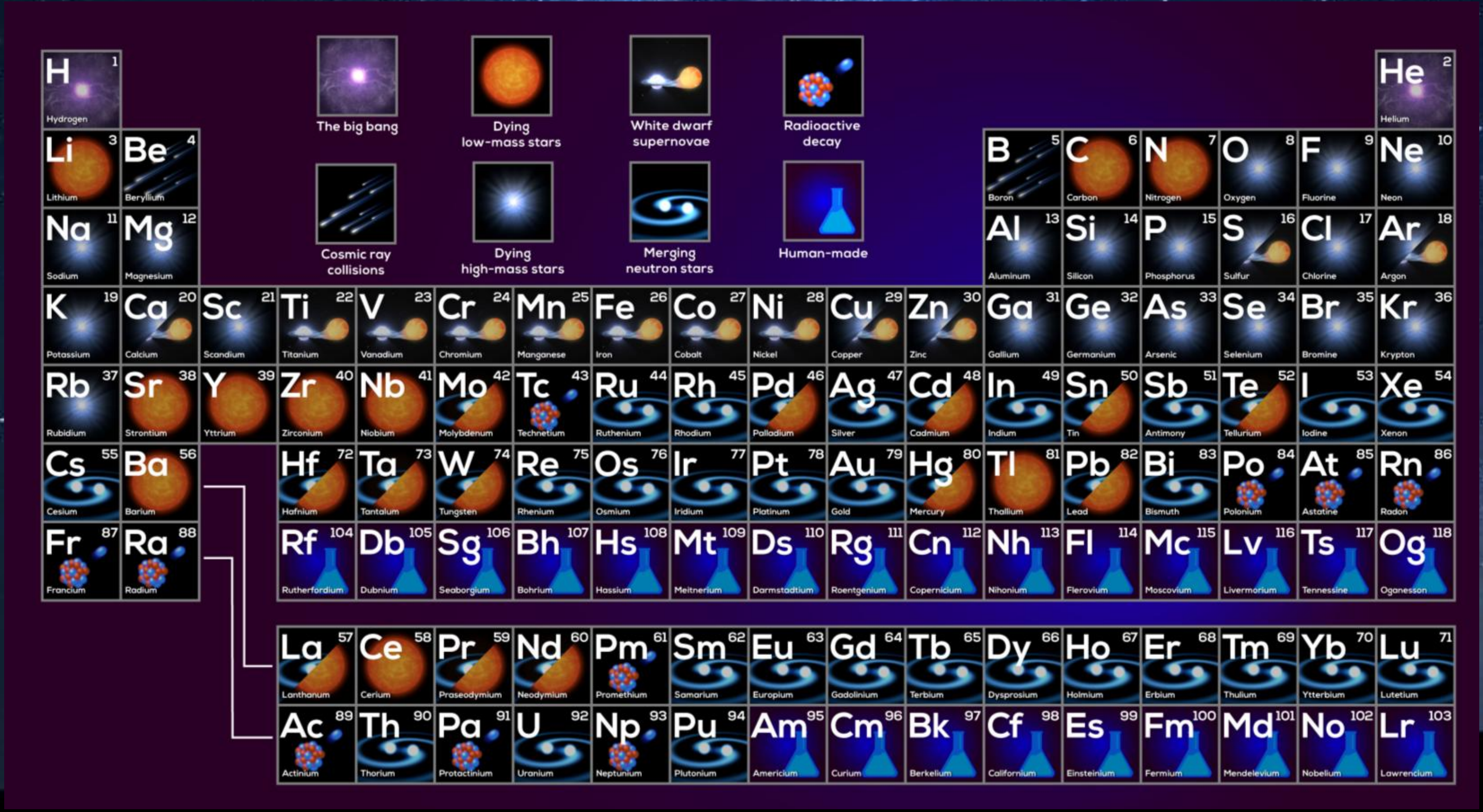
The origin of elements



[NASA's Goddard Space Flight Center]

The origin of elements

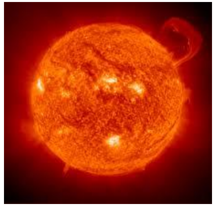


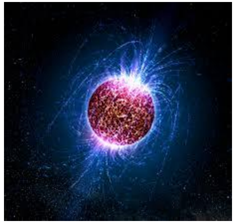
[NASA's Goddard Space Flight Center]

Density of objects in the Universe

Water  $1\ g\ cm^{-3}$

Earth  $2\ - 15\ g\ cm^{-3}$

Sun  $10 - 200\ g\ cm^{-3}$

Neutron Star  $?\ \ g\ cm^{-3}$

Density of objects in the Universe

A. $1{,}000\ g\ cm^{-3}$

B. $100{,}000\ g\ cm^{-3}$

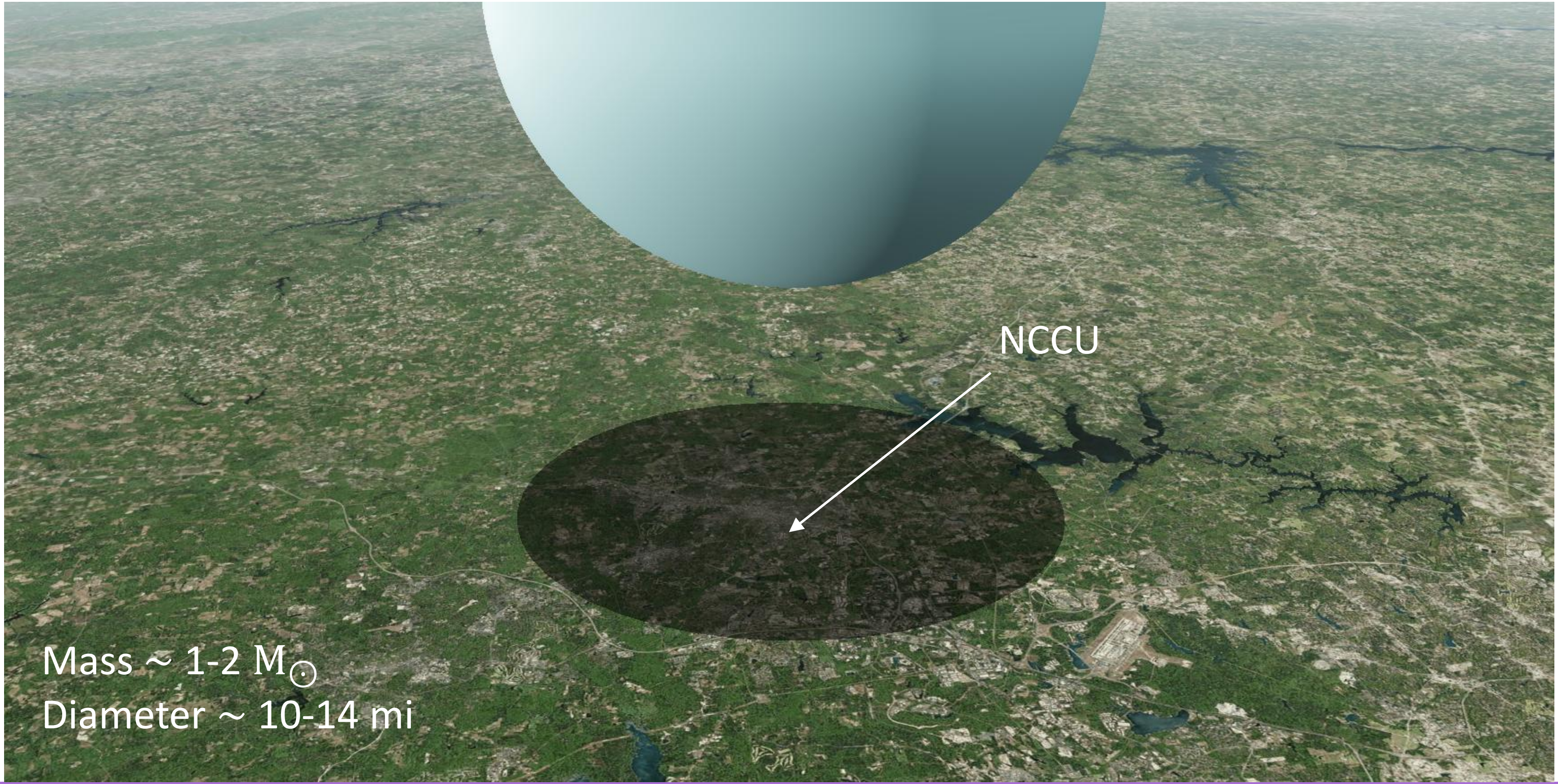
C. $10{,}000{,}000\ g\ cm^{-3}$

D. $1{,}000{,}000{,}000{,}000{,}000\ g\ cm^{-3} \sim 10^{15}$ ✓

Neutron Star

$10^{15}\ g\ cm^{-3}$

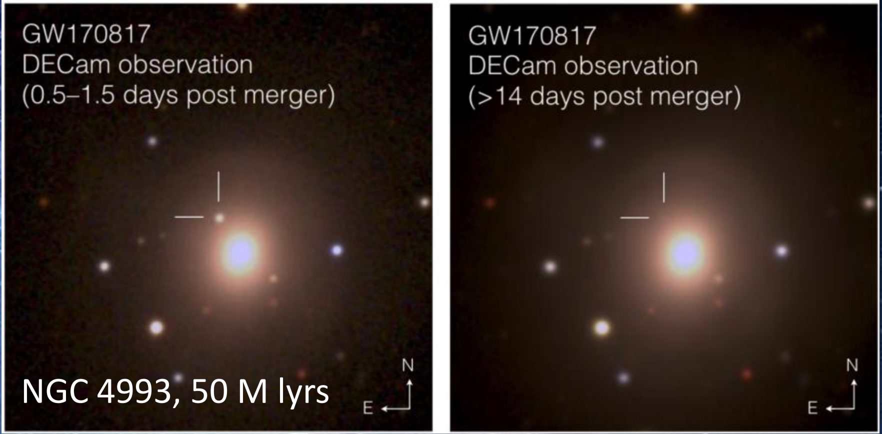
Neutron star over Durham



Neutron star merger

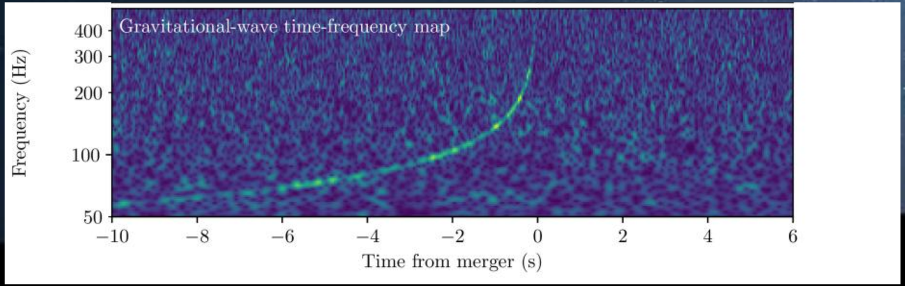
Artistic illustration from NASA Goddard

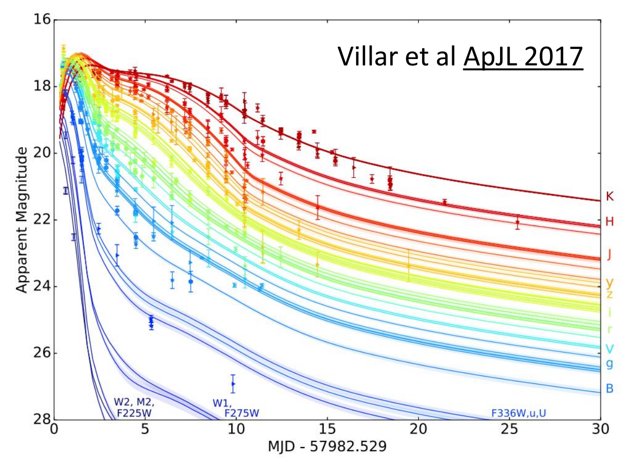
Neutron star merger: GW170817/AT2017gfo



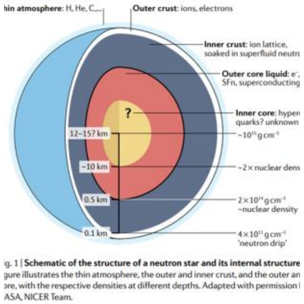
Two perpendicular 2.5 mile arms making a Michaelson-Morley setup

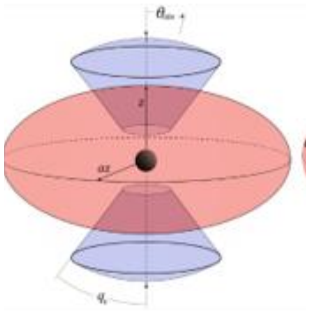
LIGO-Livingston-Louisiana, LIGO-Hanford-Washington, Virgo-Italy, KARGA-Japan, upcoming LIGO-India (2030s)

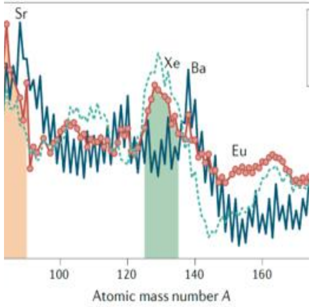




Modelling a neutron star merger







Neutron star structure

Equation of state

Bovard+17, Radice+18, Prakash+22, **Kedia**+22, **Kedia**+24, Hensh+24

Hydrodynamics

General Relativistic

Breschi+21, Bulla+19,23, Heinzel+21, Kawaguchi+20,21, **Kedia**+23, Fryer+24

Nuclear reactions

r-process nucleosynthesis

Mumpower+16, Vassh+18,+20, Zhu+18,Holmbeck+23, **Kedia**+26

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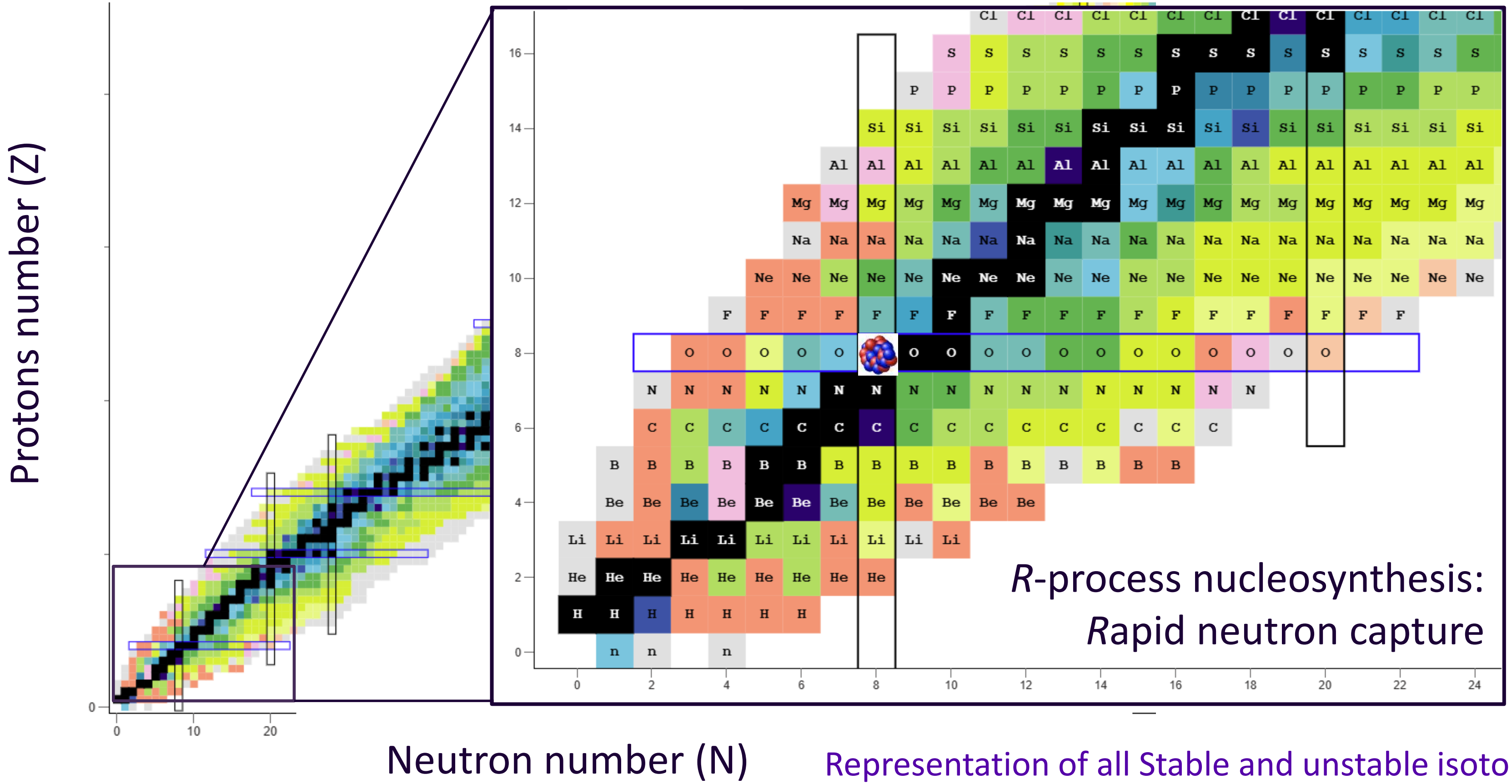
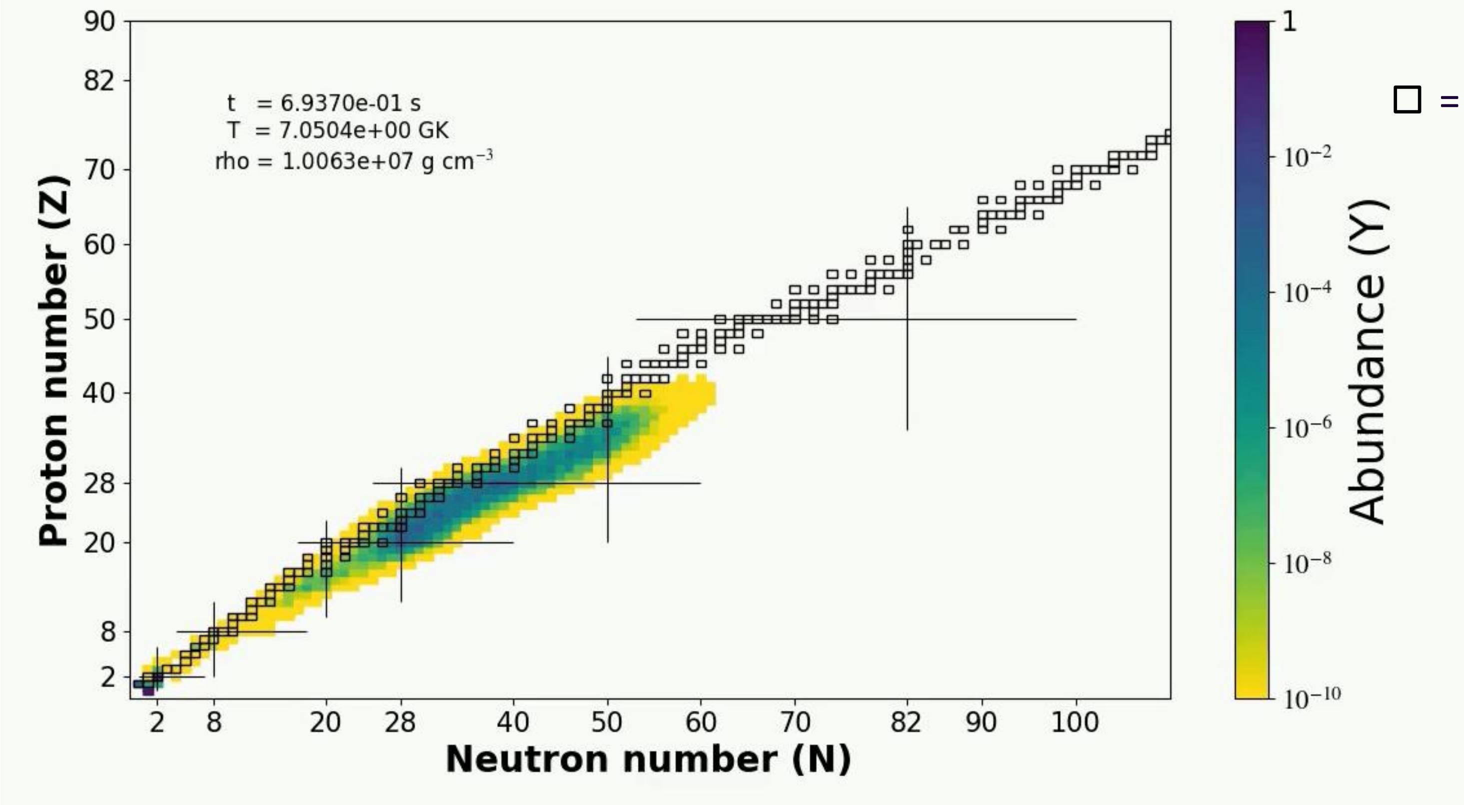
R-process on the Chart of Nuclides



Representation of all Stable and unstable isotopes

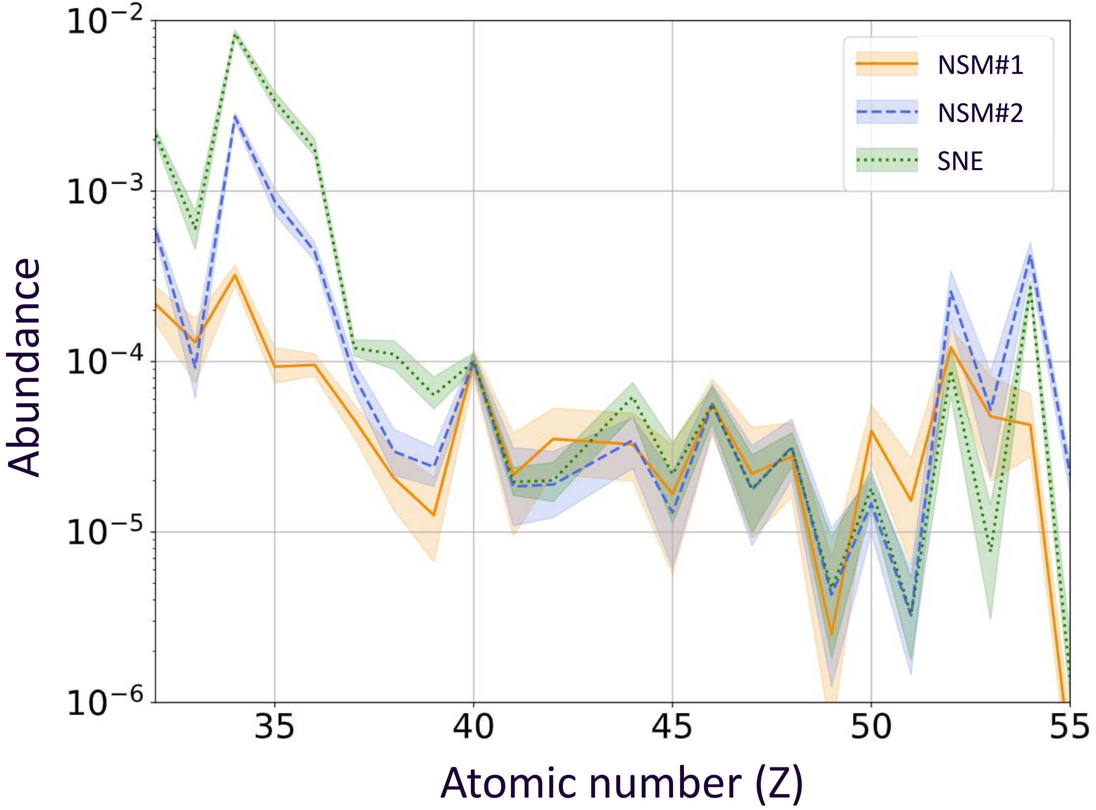
R-process evolution



☐ = Stable isotope

Movie by AK; Calculation done on PRISM (reaction network code)

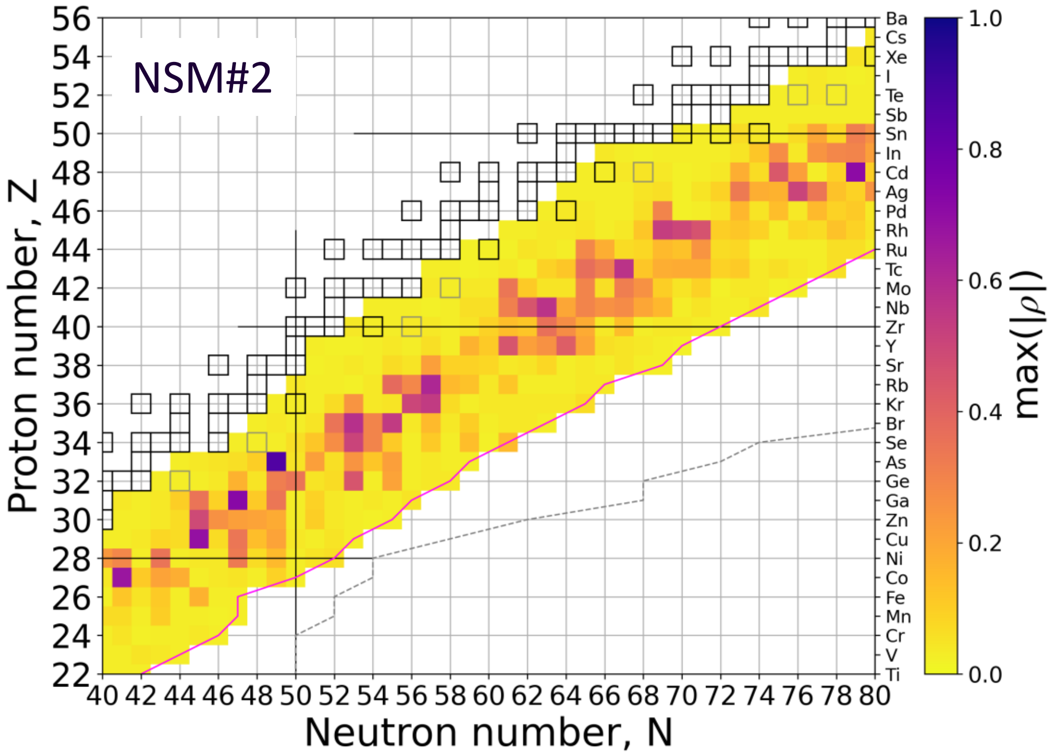
Uncertainties in *r*-process abundances



2σ (95 percentile) uncertainty in abundances due to uncertainties in theoretical neutron capture reaction rates.

*Shown is only the weak r-process abundance pattern.

Most influential isotopes for *r* process



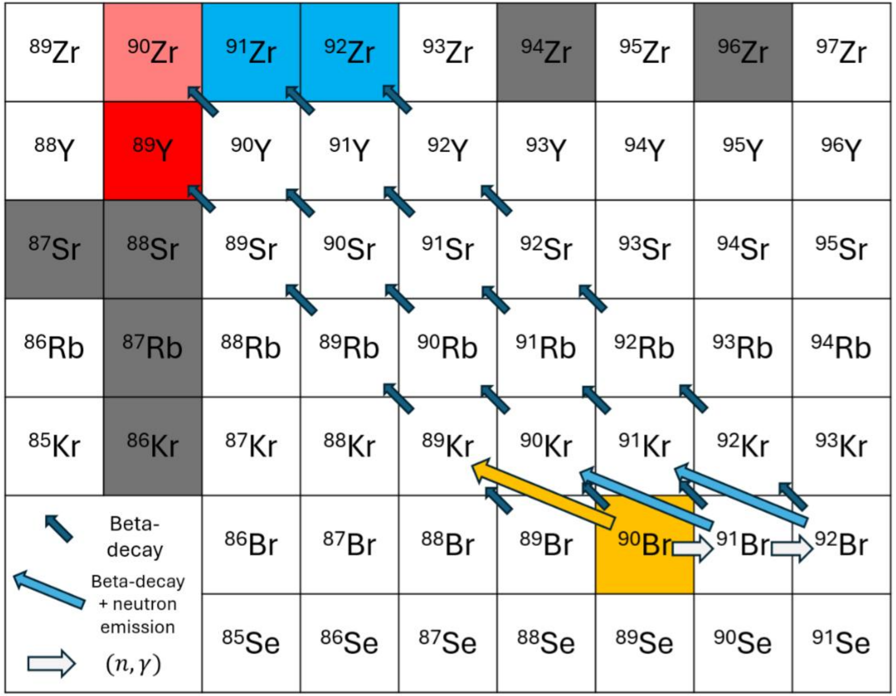
Element	(n,γ) isotope, correlation coeff. (MF14)		
Se	^{78}Ga -0.65	^{80}Ga -0.33	-
Br	^{78}Ga 0.72	^{80}Ga 0.34	^{81}Ge -0.32
Kr	^{82}As 0.86	^{82}Ge 0.35	-
Rb	^{87}Se -0.51	^{85}Ge -0.45	-
Sr	^{88}Br -0.58	^{87}Br 0.35	-
Y	^{90}Br -0.54	^{89}Br -0.30	-
Zr	^{92}Kr -0.47	^{94}Rb -0.40	-
Nb	^{93}Kr -0.54	^{92}Kr 0.50	^{92}Rb 0.38
Mo	^{94}Rb 0.61	^{100}Y -0.45	-
Ru	^{104}Nb -0.57	-	-
Rh	^{103}Nb -0.45	^{103}Y -0.42	^{103}Zr -0.36
Pd	^{110}Tc -0.47	^{104}Nb 0.34	-
Ag	^{108}Tc 0.37	^{109}Tc -0.32	^{109}Mo -0.32
Cd	^{110}Tc 0.57	-	-
In	^{114}Rh 0.53	^{115}Rh -0.50	-
Sn	^{116}Rh 0.46	^{124}Ag -0.34	-
Sb	^{123}Ag -0.51	^{123}Cd -0.44	^{120}Ag 0.33
Te	^{130}In -0.62	^{129}Cd 0.33	-
I	^{127}Cd -0.71	-	-
Xe	^{130}In 0.62	^{129}Cd -0.38	^{129}Sn -0.32

NCSU, Lawrence Livermore National Lab, Notre Dame

Kedia, et *al*., arXiv:2602.12428 (2026)

Yttrium – ^{90}Br

^{90}**Br** has a 25% probability of beta-delayed neutron emission and indirectly impact **Yttrium** formation



Kedia, et *al*., arXiv:2602.12428 (2026)

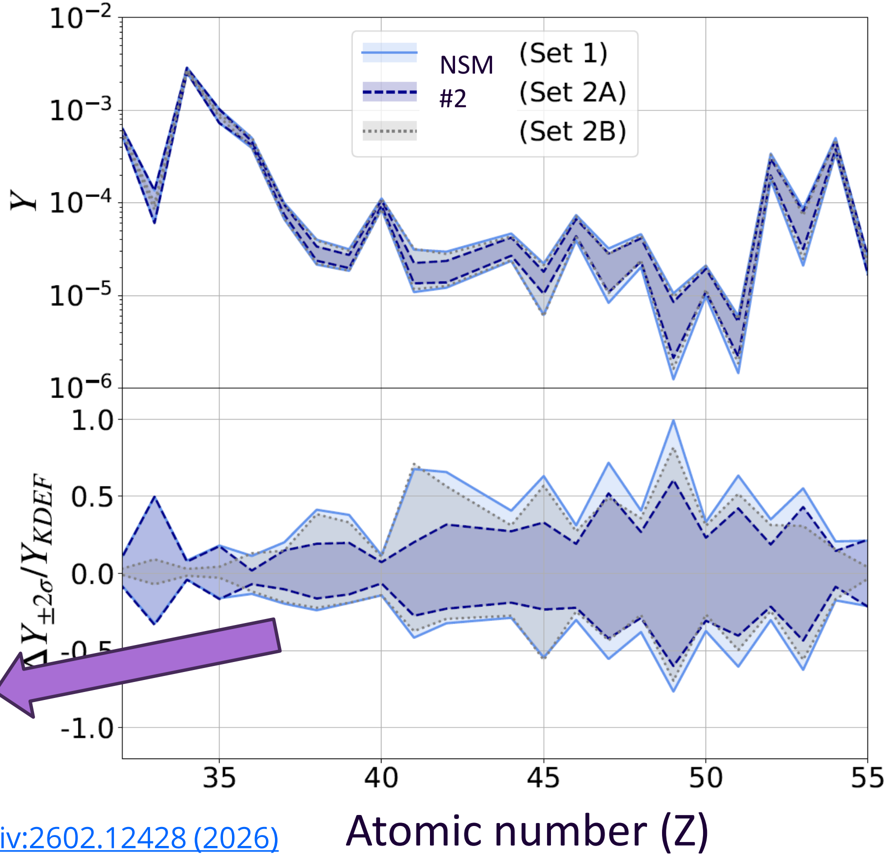
Reduction in uncertainties

2σ (95 percentile)uncertainty bands
Hypothetical reduction in uncertainties

Set 1 : No reduction

Set 2A: Reduce uncertainties on rates for only **50** isotopes

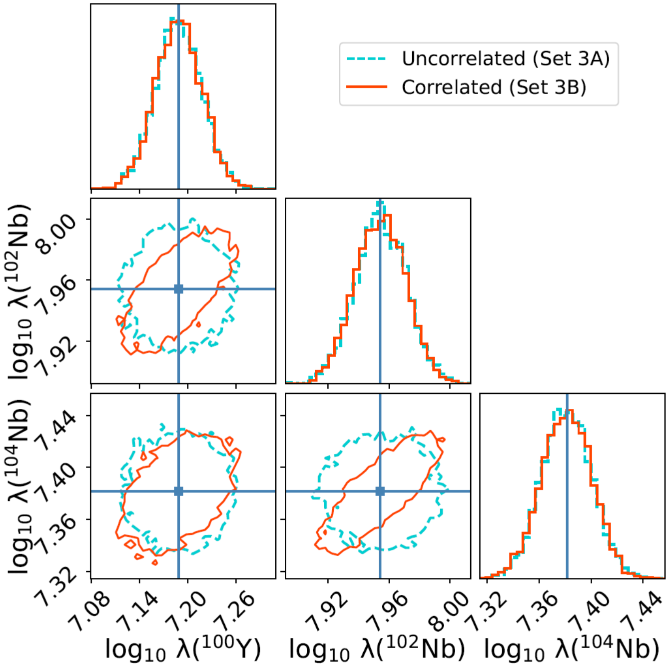
Set 2B: Reduce uncertainties on rates for all **1000** but not for the 50 isotopes.

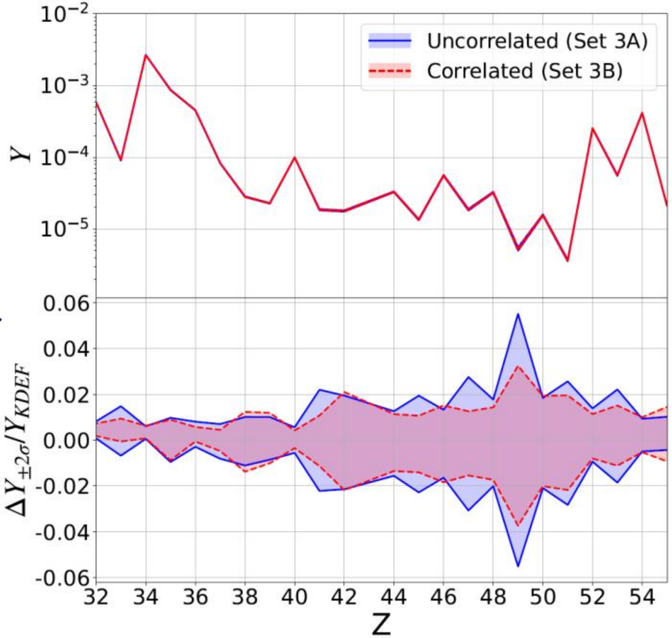


The 50 isotopes lead to more reduction in uncertainty than the remaining 1000 do!!

Kedia, et *al.*, [arXiv:2602.12428 (2026)]

Theoretical constraints of Reaction rates





Kedia, et *al*., [arXiv:2602.12428 (2026)]

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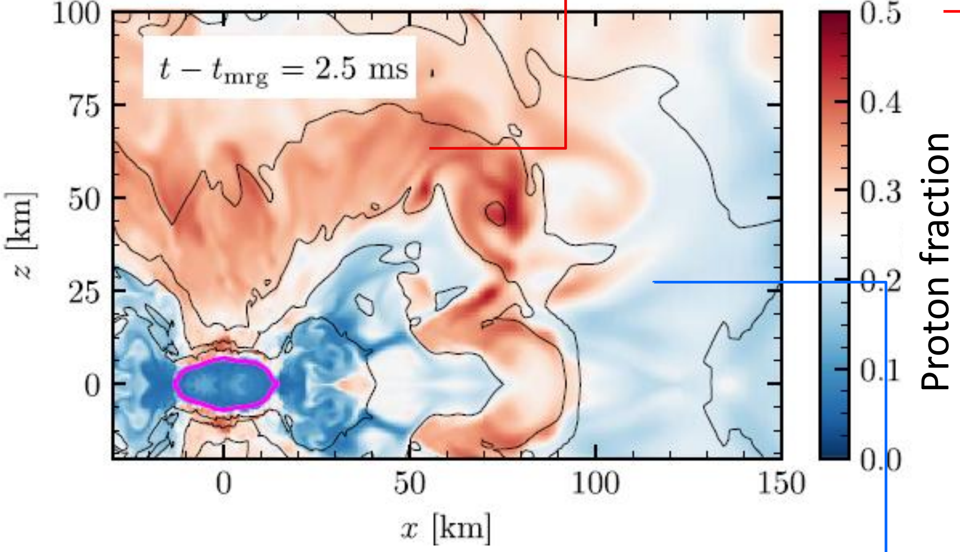
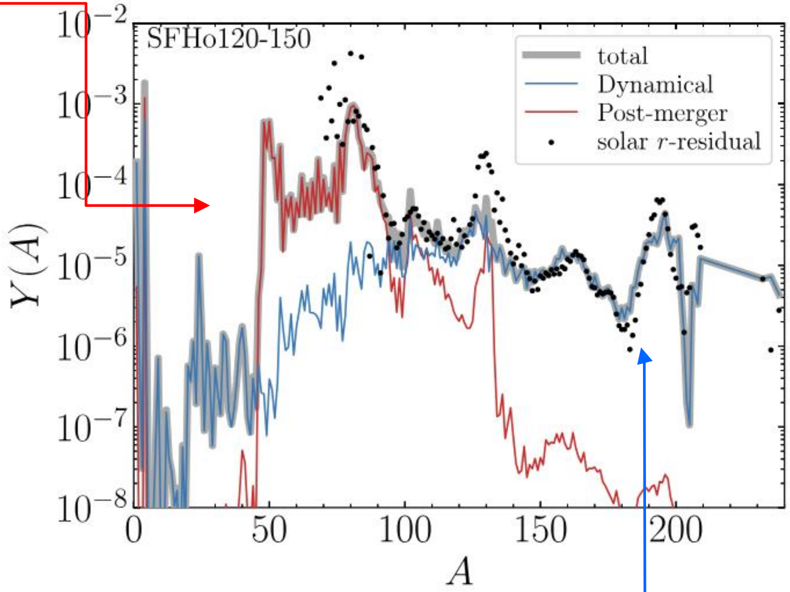
Abundance dependence on ejecta

Weak *r* process



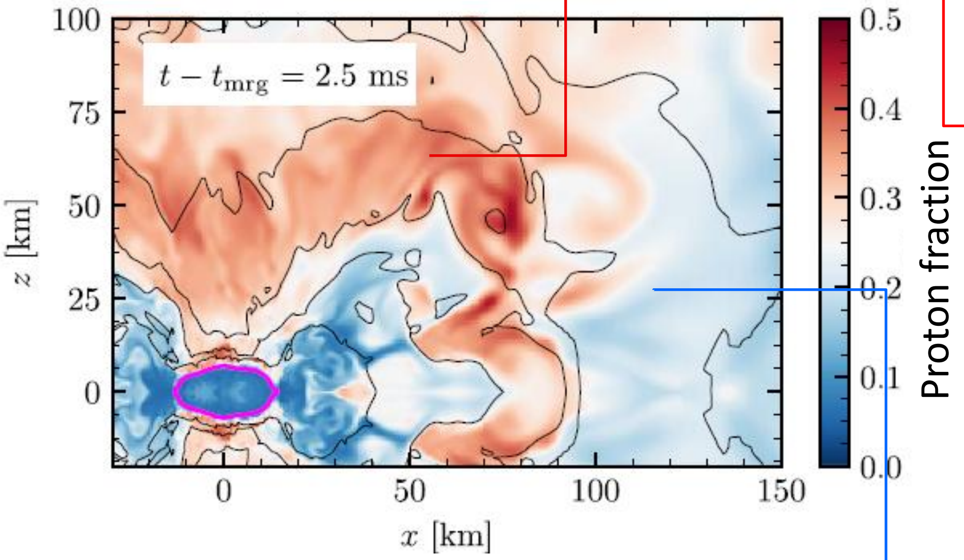


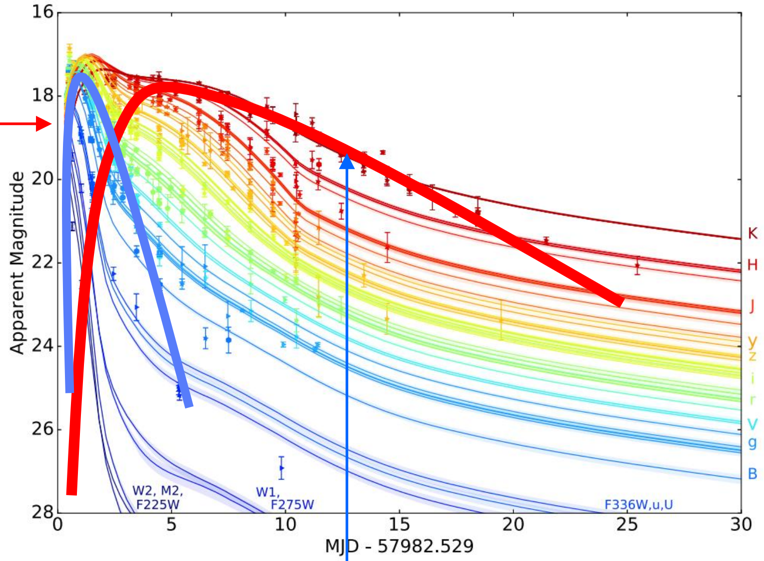
Main *r* process

$$\text{Proton fraction} = \frac{n_p}{n_p + n_n}$$

Kilonova contribution of the ejecta

Weak *r* process





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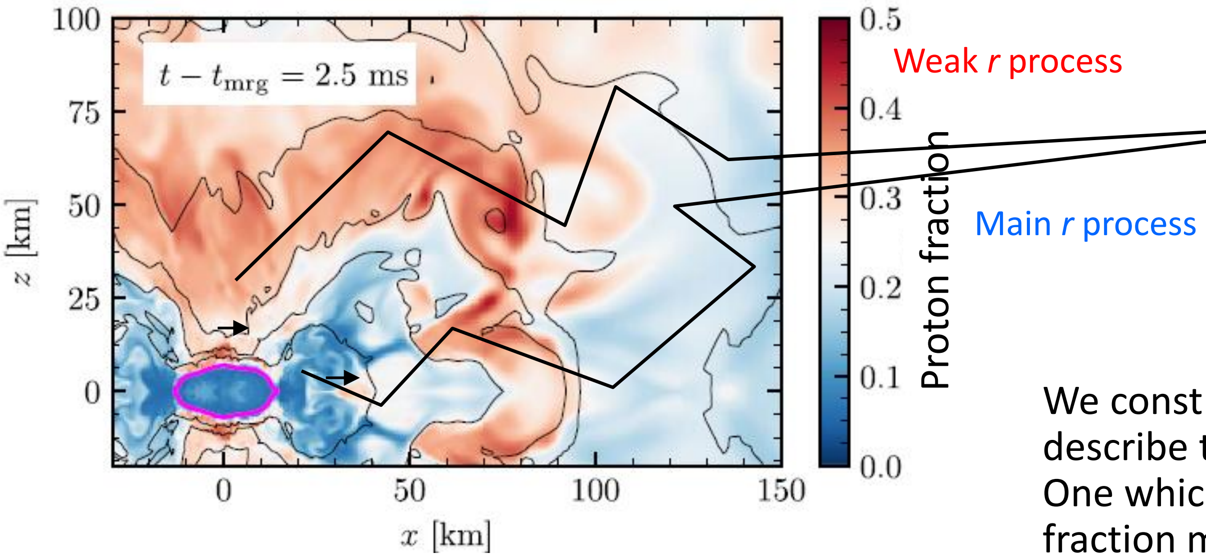
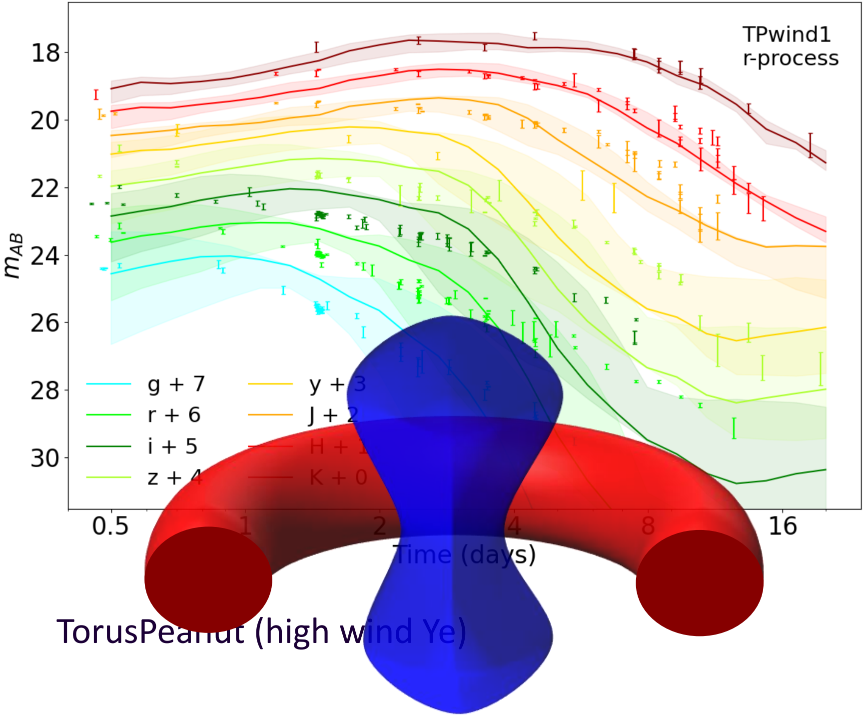


We construct a two component to describe the ejecta.
One which describes low proton fraction matter, and one which has a high proton fraction matter.

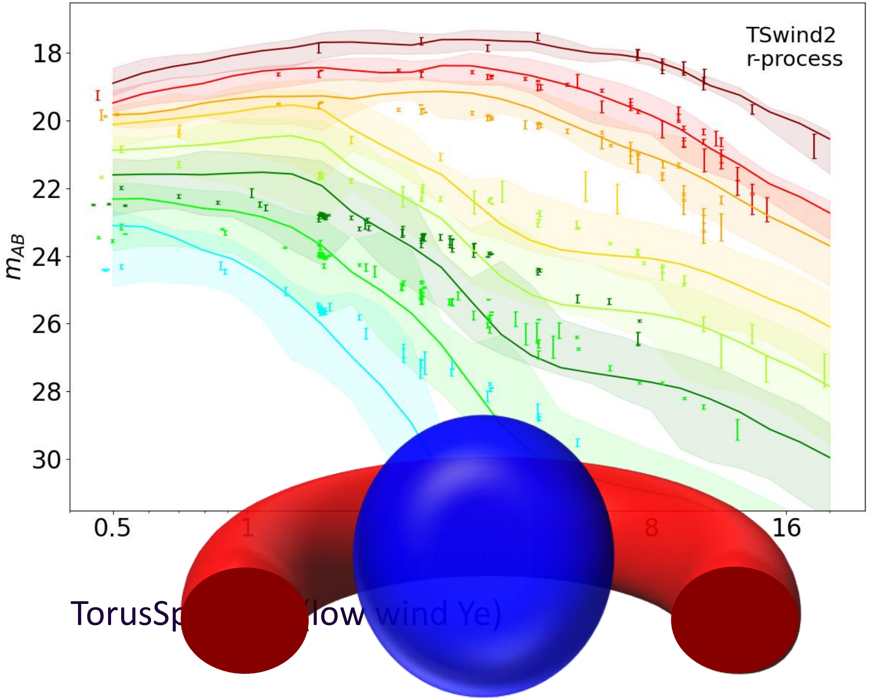
Radiative transfer model

- Radiative transfer **SuperNu**. (Wollaeger et al 2013, 2014, Wollaeger, .., Kedia, Zenodo, 2023)
- Nucleosynthesis and heating from **WinNet.** (Winteler et al. 2012, Korobkin et al. 2012)
- Thermalization efficiencies (Barnes et al. (2016))
- Atomic opacities (Fontes et at. 2020)

Kilonova Light curves



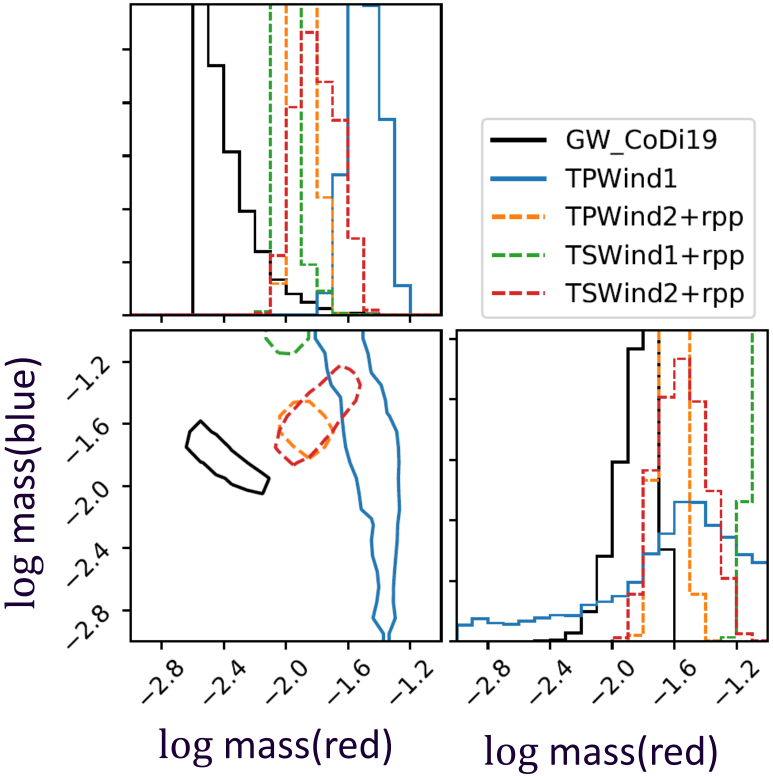
TorusPeanut (high wind Ye)



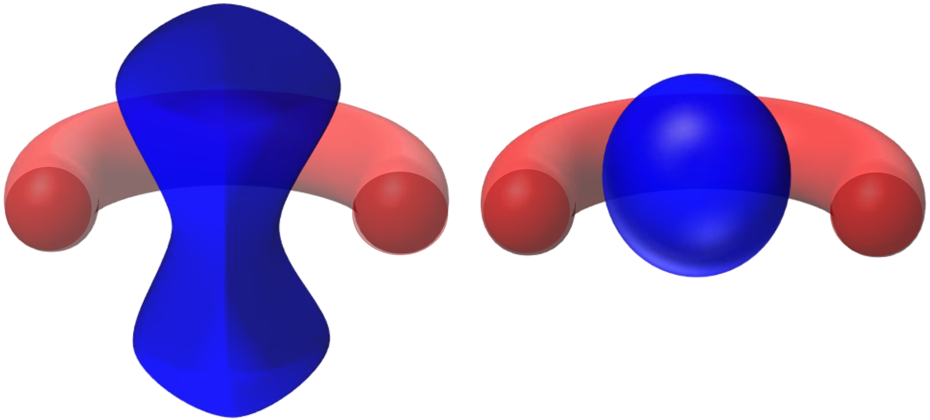
TorusSphere (low wind Ye)

AK et al Phys. Rev. Research 5, 013168 (2023)

Bayesian inference based ejecta masses from GW170817

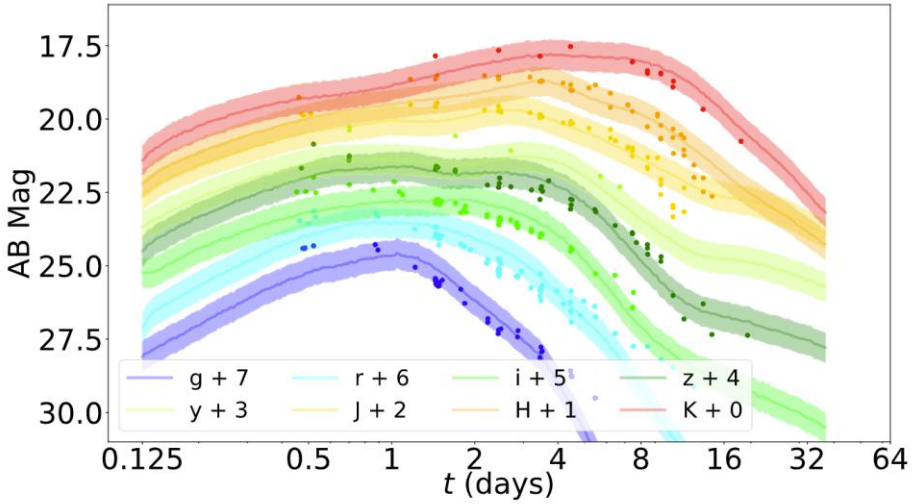


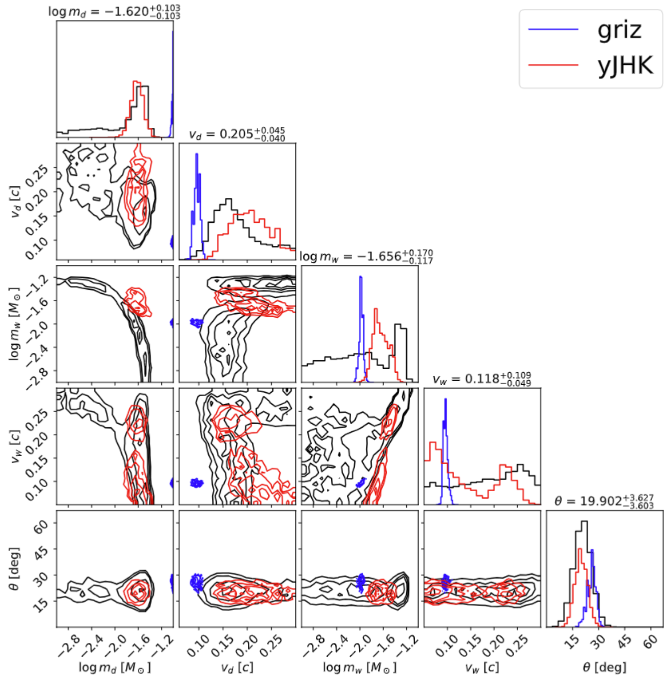
General agreement between the GW and kilonova observations on the inferred *r*-process enriched ejecta. However, how much mass was ejected dependents on the morphology.



AK, et al, Phys. Rev. Res, 5, 013168 (2023);

Light curves w/ Neural Networks





Yinglei Peng*, Ristic, Kedia, et al. [Phys. Rev. Res. 6, 033078 (2024)]
*Conducted as undergrad at URochester, currently a graduate student at GeorgiaTech

Inference dependence on chosen wavebands.

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   - Question: what is the astrophysical origin of *heavy* elements?
     - **Answer: Neutron star mergers -> Maybe**
2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)
3. Modelling Astrophysics: How much *r*-process is produced? (mass) (Answers: earlier slides)
4. Conclusions

Future of observations and experiments

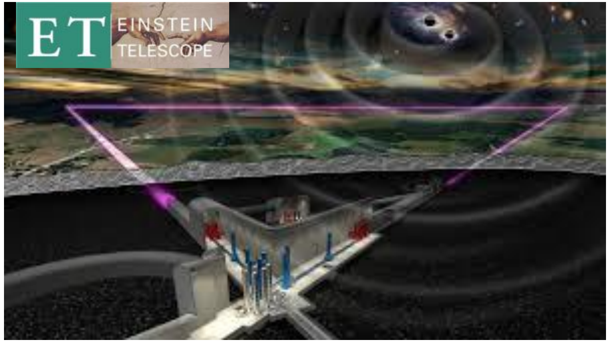
TUNL
TRIANGLE UNIVERSITIES NUCLEAR LABORATORY

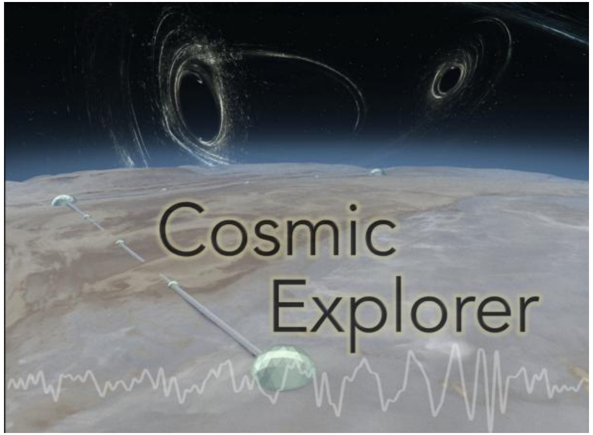
FRIB

Nuclear experiments at:
TUNL (Triangle), FRIB (MSU),
ARIEL (TRIUMF)

EM observation:
Nancy Grace Roman Space Telescope,
Vera Rubin Observatory,
James Webb Space Telescope

GW observation:
LIGO-Virgo-KAGRA
3rd Generation GW detectors





Summary

- **Origin of elements**
- **Neutron star mergers — GW170817**
  - Supernovae
- ***r*-process nucleosynthesis**
  - Uncertainties in abundance from theoretical models
  - Isotopes that reduce uncertainties
- **Kilonova modelling**
  - Modelling in the two component (red, blue) method
  - Uncertainties in the amount of ejected mass produced.

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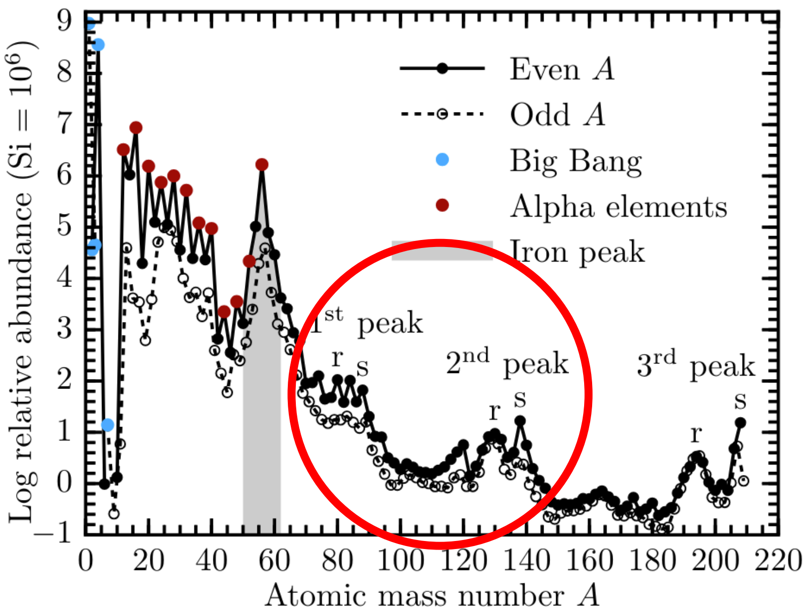
Outlook

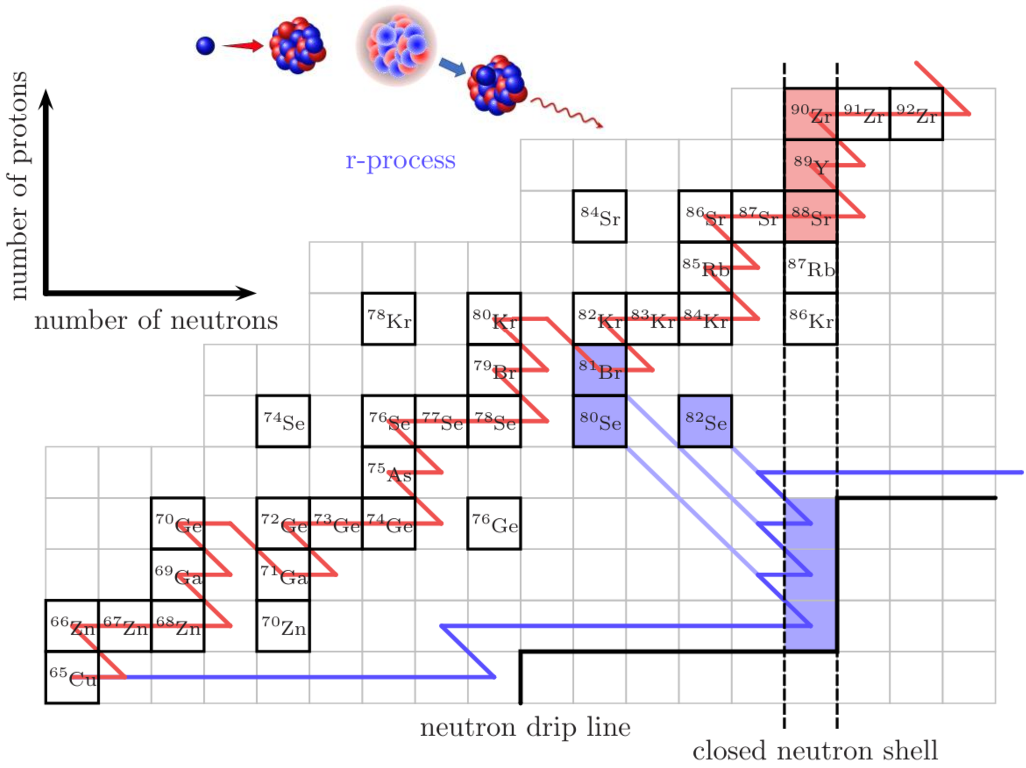
- **r-process nucleosynthesis**
  - Constraints on nuclear reaction rates
  - Astrophysical environments: Supernovae, Neutron star mergers, Collapsars, etc.
- **Kilonova modelling**
  - Apply hydrodynamic data into kilonova model
  - Model asymmetries such as due to neutron star-black hole merger.
  - Nuclear uncertainties in radioactivity

More theory, observational, and experimental work in nuclear astrophysics.

Extra slides

Solar chemical composition And the *r* process





Figures from [Lippuner (2018)]

r-process: $\tau_{n-capture} \ll \tau_{beta-decay}$

Astrophysical sites for the *r* process

Binary Neutron star mergers:
(Meyer+1989, Metzger & Fernandez 14, Just+15, Radice+18, Kedia+22, Lund+24)

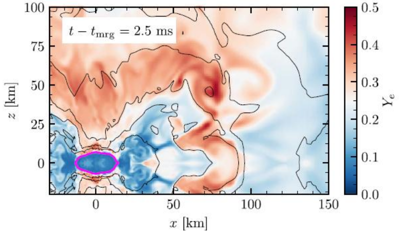
Neutron star-black hole mergers:
Surman+08, Darbha+21, Curtis+23

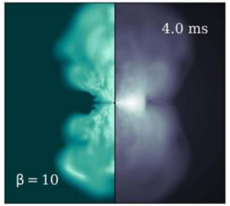
Magnetorotational supernovae:
Nishimura+15, Mosta+18, Reichert+21, Zha+24

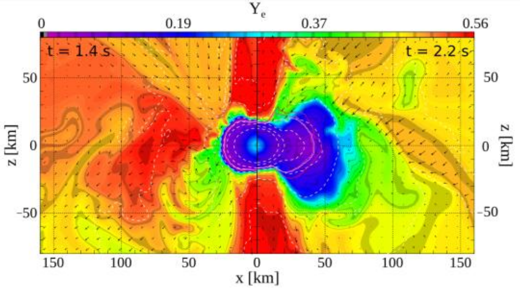
Magnetar giant flares:
Patel+25a, 25b

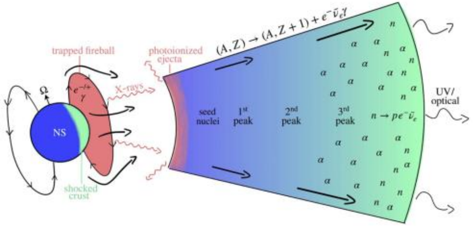
Collapsars:
Siegel+2019, Anand+2024

and **Others exotic scenarios**









Astrophysics and nuclear physics inputs

Astrophysics:

Magnetized and non-magnetized disk winds from neutron star mergers (K. Lund et al 2024, Metzger & Fernandez 2014)

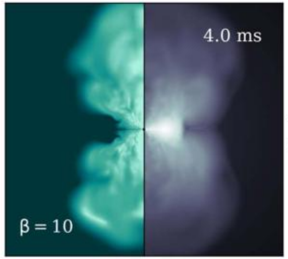
Magnetorotational supernova (Reichert et al 2021)

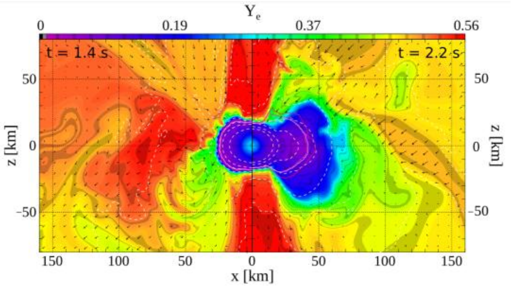
Nuclear physics:

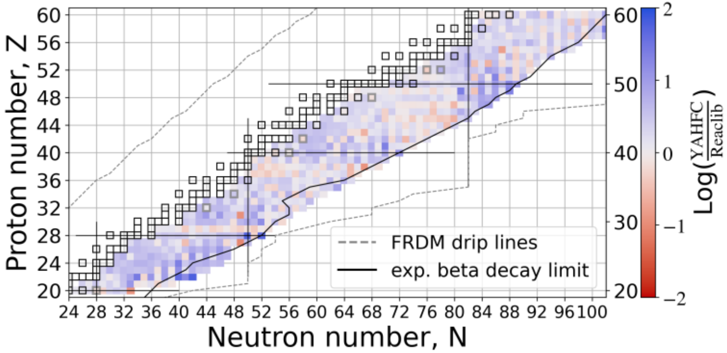
Neutron capture rates from YAHFC (Yet Another Hauser-Feshbach Code) (Ormand, Kravvaris, LLNL)

1000 isotopes, Z=20-60;

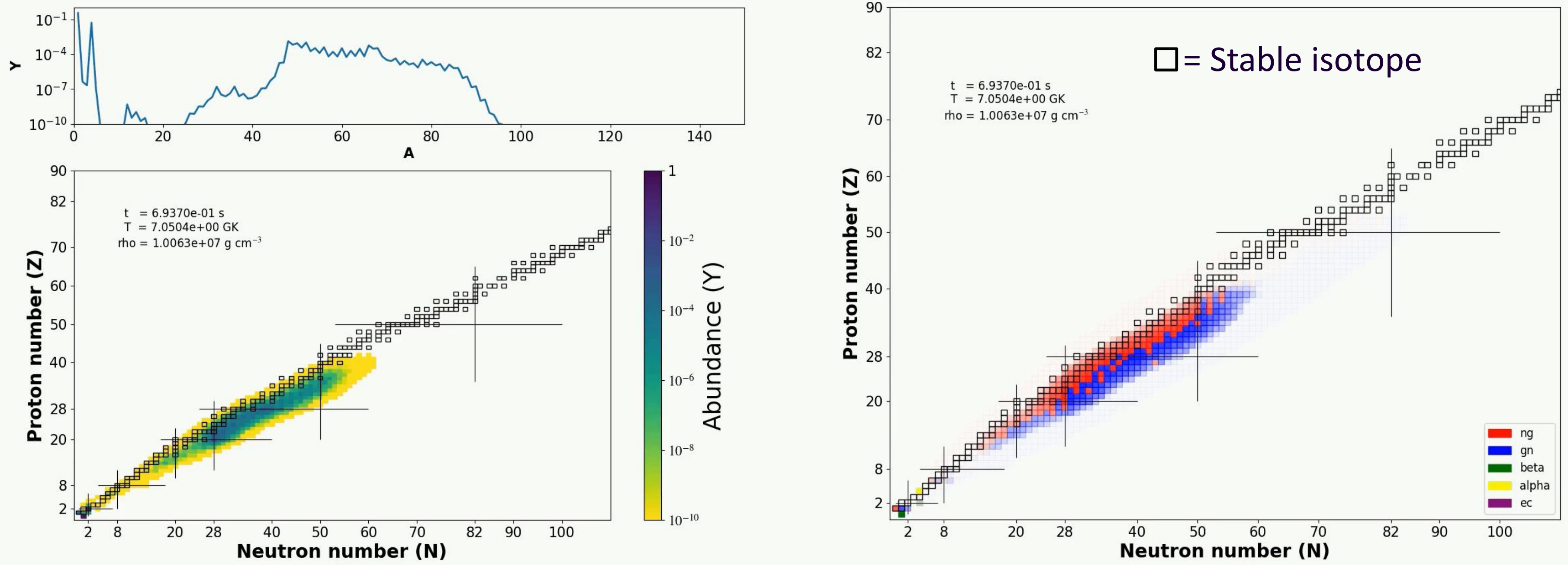
Beta-decay rates







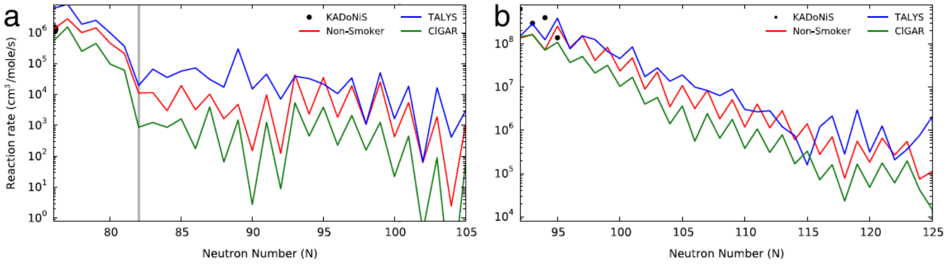
R-process evolution



ng -> neutron capture
gn -> photodissociation (neutron removal)
beta -> beta decay

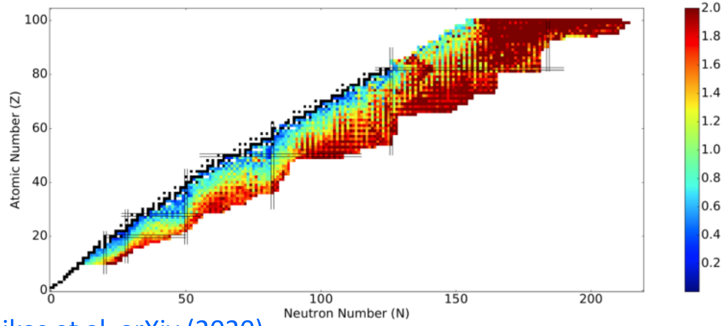
Movies by AK; Calculation done on PRISM (reaction network code)

Uncertainties in neutron capture reactions



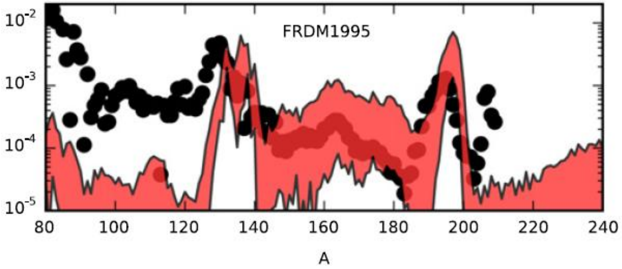
Sn –chain

Eu –chain



Nikas et al, arXiv (2020)

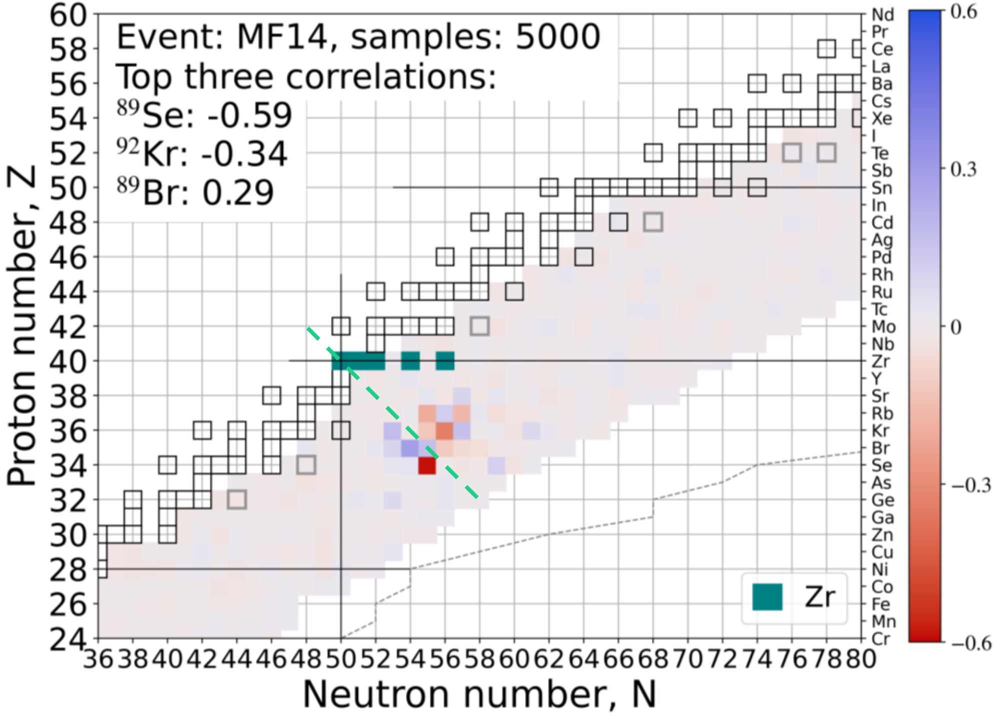
N-capture uncertainties

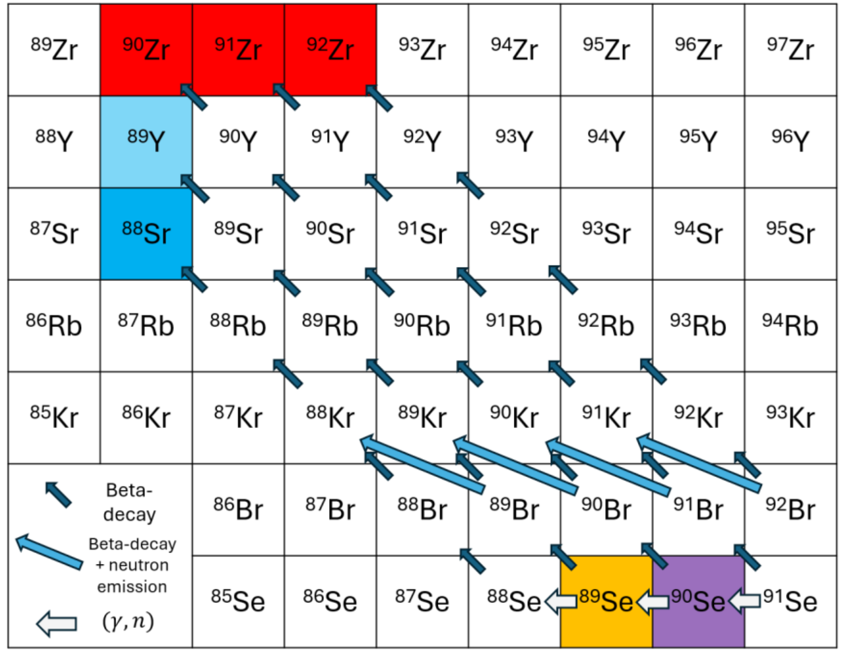


Mumpower,+, PPNP (2016) ;
multiplicative factors: $10^{-3} - 10^{3}$ in n-cap rates;
Mumpower et al ApJ 970:173 (2024)

N-cap uncertainties propagated to abundances

Elemental abundance – isotopic neutron capture rate correlation

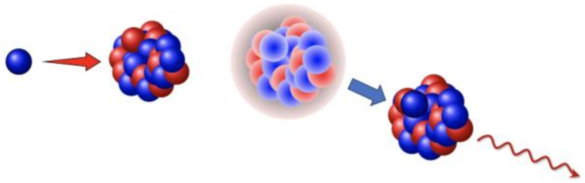




Due to detailed balance: $\lambda_{\gamma,n}^{Z,A+1} \propto \lambda_{n,\gamma}^{Z,A}$

Nomenclature: Reactions at play in *r*-process

Neutron Capture $I^A + n \;\rightarrow\; \gamma + I^{A+1}$ $I^A(n,\gamma)I^{A+1}$



Photodissociation $I^A + n \;\leftarrow\; \gamma + I^{A+1}$ $I^{A+1}(\gamma,n)I^A$

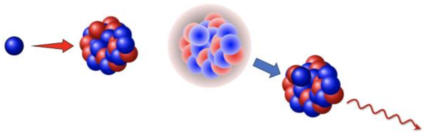
Beta-decay $I_Z^A \;\rightarrow\; I_{Z+1}^A + e^- + \overline{\nu_e} + \gamma$

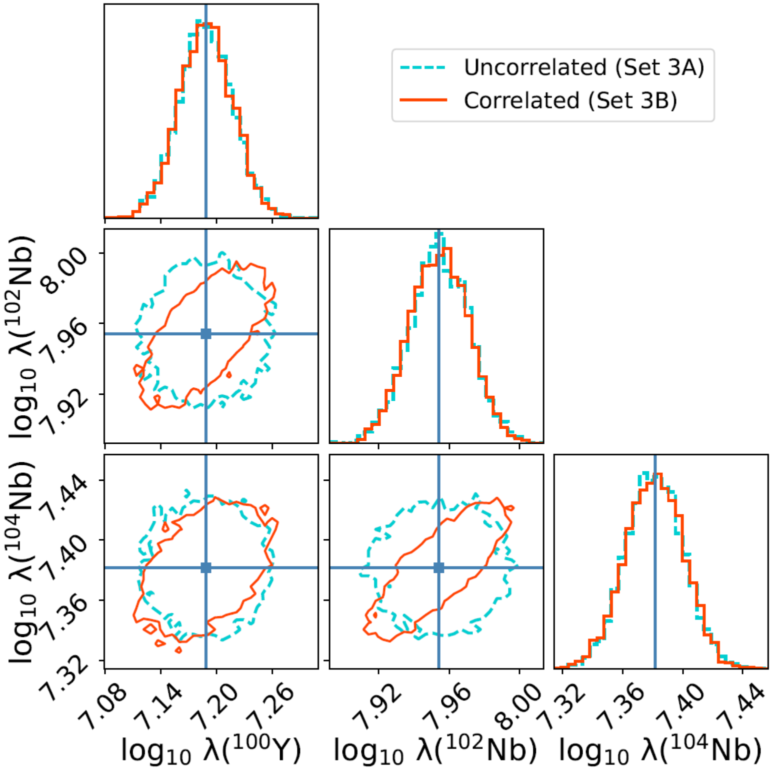
Hauser-Feshbach statistical model

Hauser-Feshbach statistical model inputs:

- ❑ **Mass Model**
- ❑ **Optical Model Potential**
- ❑ **Level Density**
- ❑ **Gamma-Strength Function**

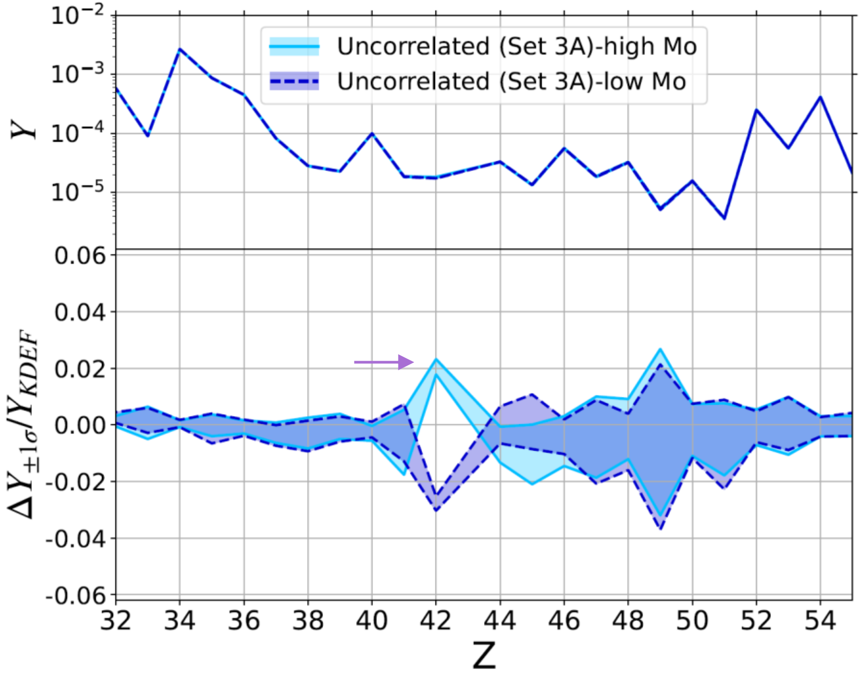
$I^{A}(n,\gamma)I^{A+1}$

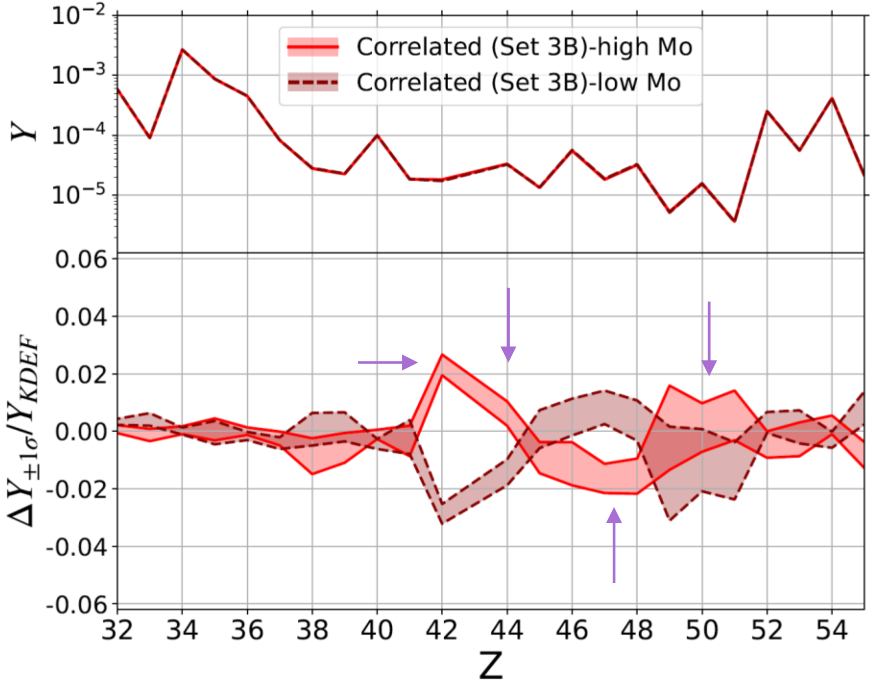




Kedia, et *al*., arXiv:2602.12428 (2026)

Correlated neutron capture rates



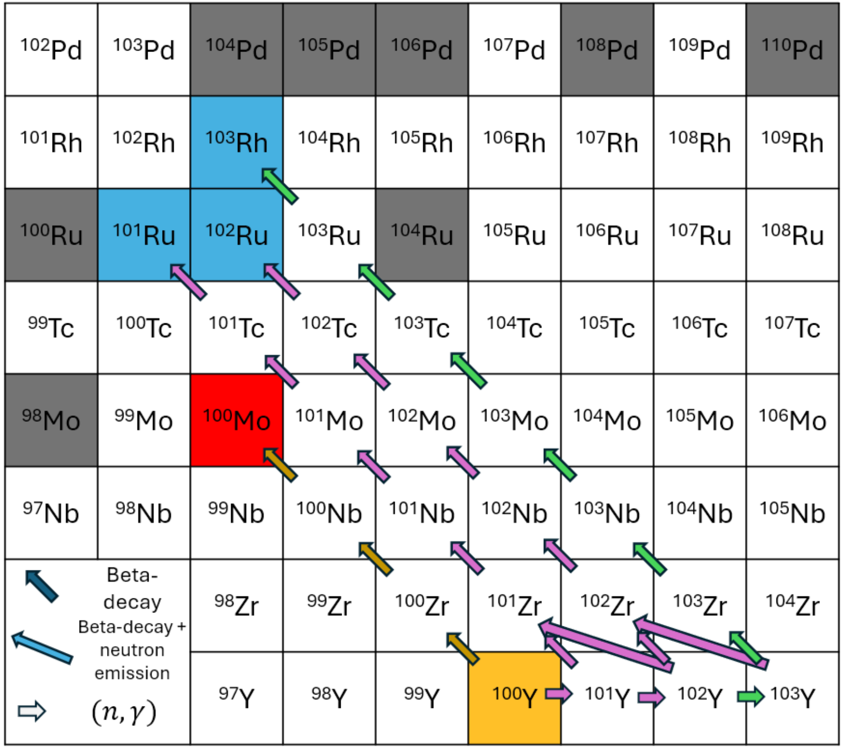


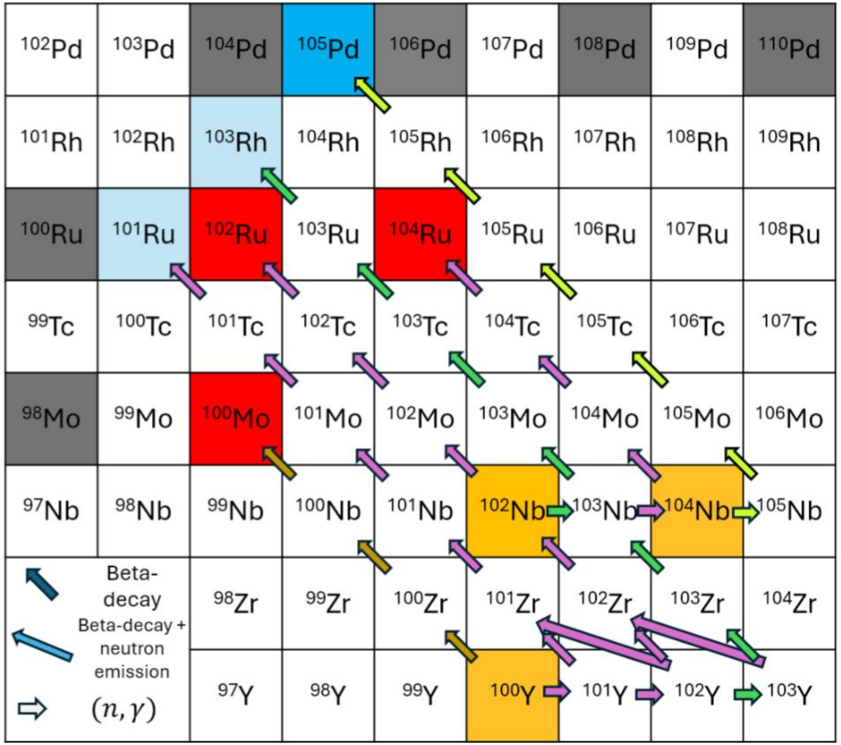
Uncorrelated: Simulations with high Mo (42) do not exhibit an increase or decrease in most other elements.

Correlated: Simulations with high Mo (42) also have high Ru (44), Sb(51) and low Rh (45), Pd (46), Ag (47), Cd (48).

Kedia, et *al*., [arXiv:2602.12428 (2026)]

Correlations in neutron capture rates – Toy model





Kedia, et *al*., arXiv:2602.12428 (2026)

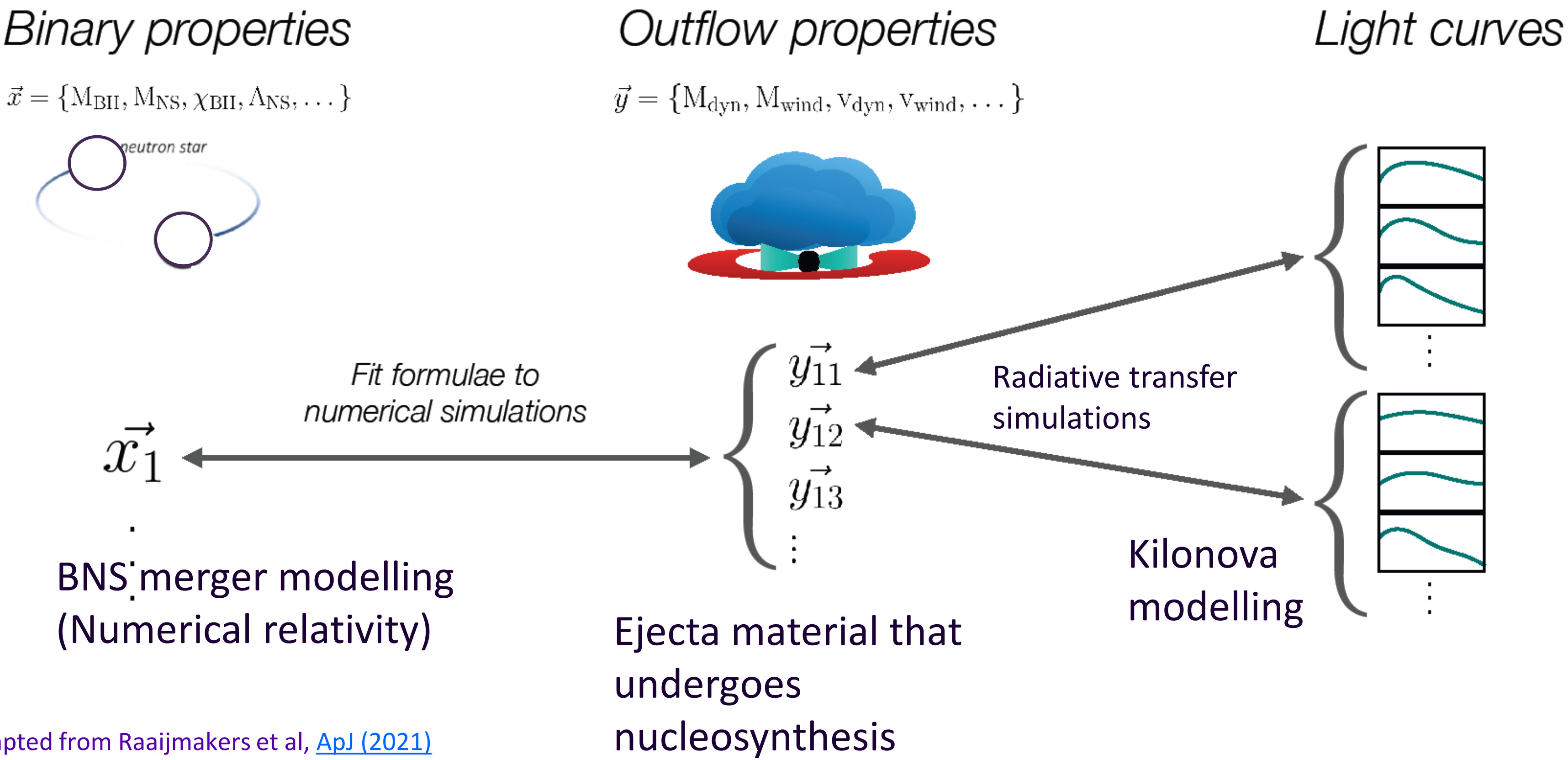
Supernova 1987a

First Multi-Messenger event

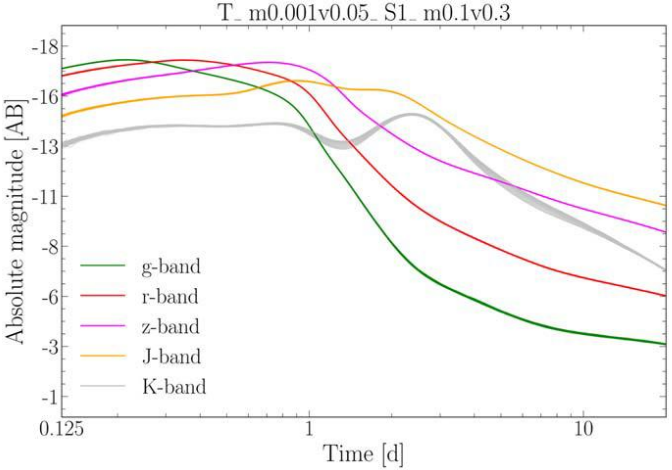
Distance ~ 50 kpc

Electromagnetic (EM) and neutrino emission detected

Bayesian inference of BNS ejecta



Adapted from Raaijmakers et al, [ApJ (2021)]

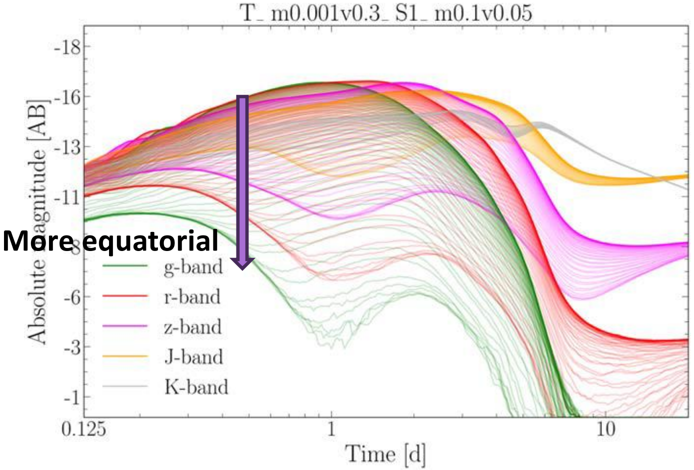


Lanthanide curtaining

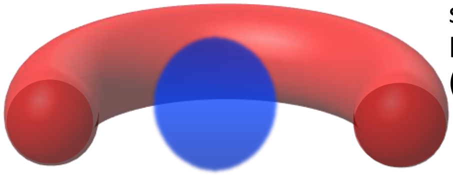
A smoking gun for r-process

Curtaining of low wavelength ("blue") light by Lanthanides (present in the "red" ejecta) produced by the nucleosynthesis.

Top panel: Ejecta more spherically symmetric and less obstructed by the "red" component. Therefore, there is virtually no angular dependence.



Bottom: "Red" component travels further and obstructs light emitted along the equator. Therefore, Lanthanide curtaining is observed. Low wavelength light is suppressed substantially along the equator. High wavelength light passes unaffected (spherically symmetric).

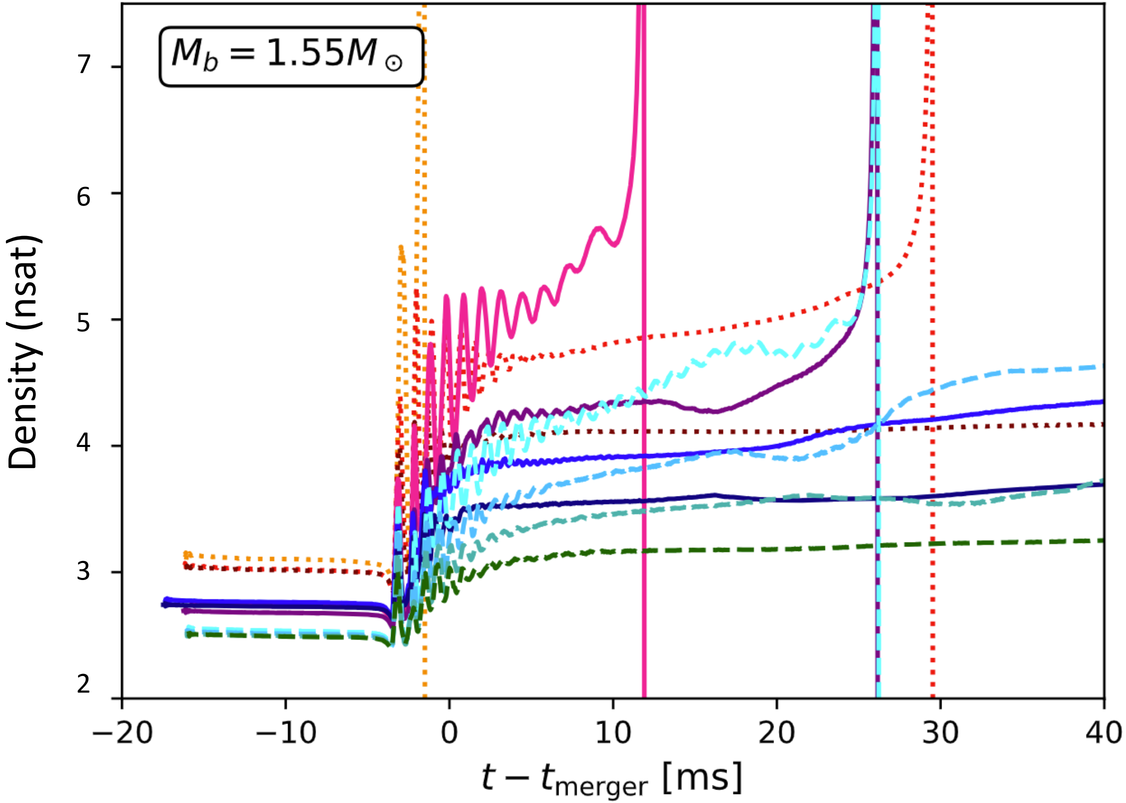


Wollaeger et al ApJ 2021; See also Korobkin+2021, Kasen+2015

Collapse of hypermassive NSs

Postmerger hypermassive Neutron star remnant dependence on the nuclear Equation of State (i.e. what defines the neutron star structure)

Vertical drop represents black hole formation.



Wei Sun, Mathews, Kedia, et al., (in prep)